2-LABORATORIYA ISHI

Mavzu: Pandas yordamida ma’lumotlarni yuklash, tozalash va tayyorlash

Maqsad

Ushbu laboratoriya ishining maqsadi:

- CSV faylni yuklash
- Ma’lumotlarni ko‘rish va tahlil qilish
- Missing value (bo‘sh qiymat)larni aniqlash va to‘ldirish
- Ma’lumotni model uchun tayyorlash
- Train-test ga ajratish

Kerakli vositalar

- Python 3.10+
- Pandas
- Scikit-learn

O‘rnatish:

```
pip install pandas scikit-learn
```

Nazariy asos

Pandas — ma’lumotlarni jadval ko‘rinishida boshqarish uchun ishlatiladi.
AI model o‘qitishdan oldin ma’lumotni tozalash majburiy bosqich hisoblanadi.

DataFrame tuzilishi

pandas

```python
df['State'].value_counts(normalize=True)
```

```
CA    0.121532
FL    0.093791
PA    0.076618
OH    0.046235
MO    0.042272
MI    0.036988
GA    0.034346
NY    0.033025
NC    0.030383
CO    0.025099
```

sidetable

```python
df.stb.freq(['State'])
```

	State	Count	Percent	Cumulative Count	Cumulative Percent
0	CA	92	0.121532	92	0.121532
1	FL	71	0.093791	163	0.215324
2	PA	58	0.076618	221	0.291942
3	OH	35	0.046235	256	0.338177
4	MO	32	0.042272	288	0.380449
5	MI	28	0.036988	316	0.417437
6	GA	26	0.034346	342	0.451783
7	NY	25	0.033025	367	0.484808
8	NC	23	0.030383	390	0.515192
9	SC	19	0.025099	409	0.540291
10	MN	19	0.025099	428	0.565390
11	CO	19	0.025099	447	0.590489

amphitheaters.csv - Notepad

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```
"Roman","Modern","Country","Year","Length","Notes","Photo","Latitude","Longi
"Dyrrhachium","Durrës","Albania","2nd century AD","61 m","Durrës Amphitheatr
"Lambaesis","Lambèse","Algeria","","64 m","","",35.489247,6.259935
"Colonia Claudia Caesarea","Cherchell","Algeria","","93 m","","",36.60874,2.
"Gemellae","M'lili","Algeria","","37 m","","",34.635409,5.522764
"Theveste","Tébessa","Algeria","4th century AD","45 m","Aerial Photograph","
"Tipasa","Tipaza","Algeria","","","Map of Tipasa","https://en.wikipedia.org/
"Carnuntum","Petronell","Austria","","69 m","2 amphitheatres ","https://en.w
"Carnuntum","Petronell","Austria","","69 m","2 amphitheatres ","https://en.w
"Flavia Solva","Leibnitz","Austria","","","","",46.766744,15.567417
"Virunum","Magdalensberg","Austria","","","","https://en.wikipedia.org/wiki/
"Diocletianopolis","Hisarya","Bulgaria","","","","",42.502825,24.709776
"Marcianopolis","Devnya","Bulgaria","","","","",43.222222,27.569444
"Serdica","Sofia","Bulgaria","3rd century AD","","In ground floor of Arena c
"Pietas Iulia Pola","Pula","Croatia","1st century AD","68 m","Pula Arena","h
"Salonae","Solin","Croatia","","65 m","","https://en.wikipedia.org/wiki/File
"Burnum","","Croatia","","46 m","Roman military camp near Šibenik, had a sma
"Augusta Paphus","Paphos","Cyprus","","65 m","","",34.754942,32.405344
"Salamis","","Cyprus","","","Amphitheatre almost vanished ","",35.185522,33
```

ID	Color	Weight	Broken	Class
1	Black	80	Yes	1
2	Yellow	100	No	2
3	Yellow	120	Yes	2
4	Blue	90	No	2
5	Blue	85	No	2
6	?	60	No	1
7	Yellow	100	?	2
8	?	40	?	1

Ishlash tartibi (Step-by-Step)

1-qadam: Namunaviy dataset yaratish

Talabalar quyidagi CSV faylni yaratadi (students.csv):

```
math,english,physics,gender,result
85,78,90,M,1
90,,88,F,1
70,65,72,M,0
88,90,,F,1
60,55,58,M,0
```

2-qadam: CSV yuklash

```
import pandas as pd

df = pd.read_csv("students.csv")

print(df.head())
```

3-qadam: Dataset haqida ma’lumot olish

```
print(df.info())
print(df.describe())
```

4-qadam: Missing value aniqlash

```
print(df.isnull().sum())
```

5-qadam: Missing value to‘ldirish

Numerik ustunlarni o‘rtacha bilan to‘ldirish:

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

6-qadam: Gender ustunini raqamlashtirish

```
df["gender"] = df["gender"].map({"M":0, "F":1})
```

7-qadam: Model uchun X va y ajratish

```
X = df.drop("result", axis=1)
y = df["result"]
```

8-qadam: Train-Test ga ajratish

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Train hajmi:", X_train.shape)
print("Test hajmi:", X_test.shape)
```

To‘liq kod (Yakuniy variant)

```
import pandas as pd
from sklearn.model_selection import train_test_split

df = pd.read_csv("students.csv")

print(df.head())
print(df.info())

print("Missing values:")
print(df.isnull().sum())

df.fillna(df.mean(numeric_only=True), inplace=True)

df["gender"] = df["gender"].map({"M":0, "F":1})

X = df.drop("result", axis=1)
y = df["result"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Train hajmi:", X_train.shape)
print("Test hajmi:", X_test.shape)
```

Tekshirish bosqichi

Talaba quyidagilarni tekshiradi:

✅ Dataset yuklandi
✅ Missing value yo‘q
✅ Gender ustuni raqamlashtirildi
✅ X va y ajratildi
✅ Train-test muvaffaqiyatli bo‘lindi

Hisobot topshirish talabi

Talaba topshiradi:

1. Python kodi
2. Tozalangan dataset
3. Skrinshot (train-test hajmi)
4. 6-8 qatordan iborat tahlil:
    - Qancha missing value bor edi?
    - Qanday to‘ldirildi?
    - Train va test hajmi qancha?

Mezoni	**Ball**
Kod ishlashi	40
Tozalash to‘g‘riligi	25
Encoding	15
Train-test ajratish	10
Tahlil yozish	10